NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Islamabad Campus

Design and Analysis of Algorithms | BS Data Science & Artificial Intelligence

PROJECT

Total: 100 marks (+10 bonus)

LIFELINE

Algorithmic Relief-Command Engine

Submission rules

- Group. Work in a group of exactly 3 members from the same section. All three must contribute to data work, algorithm selection, implementation, experiments and analysis, and every member must be able to explain and modify the complete project.
- Project structure. This is one integrated case study to be completed over approximately two months. It covers data generation, data understanding, justified algorithm selection, implementation, testing, experiments and an integrated relief-command engine.
- Data submission:

  Submit one ZIP per group, named **RollNoSection1_RollNoSection2_RollNoSection3,** containing cases.xlsx (exactly 50 cases, excel sheet), the generator/ that writes it from the group seed, the validator/ with validation_report.txt, approach.md and members.txt. The data must come from the group's own generator; do not submit hand-typed cases, copied data or only screenshots.
- Algorithm selection. No specific algorithms are prescribed in this statement. You must inspect the generated data, identify the computational problems, research and select suitable algorithms yourselves, and justify each selection using assumptions, expected complexity, data characteristics and trade-offs. Do not simply list algorithm names.
- Submission. One member uploads one complete submission per group to GCR (Google Classroom).The submission includes source code, tests, README, the master dataset used, results, plots/tables and repository history.
- Naming Use **RollNoSection1_RollNoSection2_RollNoSection3**, for example **22I-3105-A_22I-3106-B_22I-3107-C**, for the final ZIP and repository.
- Late submissions will not be accepted.
- Evidence. All results must be reproducible. Every graph must have labelled axes with units, a legend and a caption. Explain any difference between your predictions and measurements.
- Scope. Use algorithms implemented by your group. Standard libraries may be used for data handling and verification, but must not silently perform the core algorithm being evaluated. Any permitted library use must be documented.
- Viva. Each member is examined individually. Group marks are shared, but a member who cannot explain the dataset, algorithm choices, code or results may lose individual marks.
- Academic integrity. Do not copy data, code, analysis or results from other groups. Cite references. AI tools may assist, but each member remains responsible for every submitted line and claim.
- **Vague logs.** A submission with an empty or vague Prediction Log or Recommendation (Problem N) receives no more than 40 percent of the total marks.
- **Plagiarism** results in a zero for the entire project category. Sharing solutions across groups or sections, or submitting results that you have not verified by running your own code, is handled according to the course policy.

Iteration 1: Data Generation and Algorithm Selection

In Iteration 1 every group submits two deliverables:

- **The data (Sections 1 to 5).** 50 cases of the LIFELINE world, built from six case studies with the group's own generator and checked with the group's own validator, delivered as cases.xlsx.
- **The algorithm-selection report (Section 6).** A two-page report, also called the proposal, in which the group selects and justifies the algorithms it will implement in Iteration 2 and states the contribution of each member.

The relief problems themselves are solved in Iteration 2. Both deliverables, together with the GitHub repository (Section 7), are submitted in one ZIP (Section 8).

1. Rules for the data

- The group seed **s** is the group number from the group-selection sheet. All randomness comes from s and the case number. The case_id of case k is s, a hyphen and k as two digits (for example 17-07).
- Cases 00 to 19 are **tier S** (small, an exhaustive search must be possible), 20 to 39 **tier M**, 40 to 45 **tier L**, and 46 to 49 **tier X**, the trap cases of Section 2. Tier X uses the ranges of tier L unless a trap says otherwise.
- Case k, for k = 0 to 45, belongs to **case study (k mod 6) + 1**. Its **signature rules** (below) must hold.
- All data comes from your generator. No external data, no hand-typed cases, no data from another group. All data is synthetic.
- Numbers are integers and text is plain ASCII. No formulas, merged cells, empty rows or extra sheets. Write text cells as text.

The six case studies

Each story says what the sector is like. The signature rules are the only numerical constraints it puts on you. All other numbers (sizes, weights, costs, priorities) are your decision within the ranges of Section 3, and you must defend them in your approach note.

Case study	Story	Signature rules
1. Delta Flood	A river has burst its embankments. A flat delta of small villages joined by raised roads. Roads go under water in waves. Food and water matter most	Average degree (2E / V) 2.2 to 3.2. At least 40% of locations have penalty 2 or more. At least 3 waves. Food and water crates (cat 0, 1) at least 55%
2. Mountain Earthquake	A valley town with a few roads and almost no alternatives. The injured need medicine and tools. Bursts of urgent calls follow each aftershock	Average degree 2.0 to 2.6. Medicine and tool crates (cat 2, 3) at least 45%. Mean alarm priority at least 6.0
3. Coastal Cyclone	A port city and fishing villages. Storm surge closes roads in waves. Rescue windows are short and overlap. The radio is full of noise	At least 25% of locations are shelters. At least 3 waves. Every window overlaps at least 2 other windows on average. The log uses at least 30 of the 38 symbols
4. Urban Flash Flood	A dense city with many alternative routes. Calls arrive faster than anyone can answer them. Rare equipment is available	Average degree at least 3.5. On tiers M and L at least 1.5 alarms per tick. Rare crates (cat 4) at least 10%
5. Landslide Corridor	A single mountain highway along a gorge. A collapse cuts off whole communities	Average degree at most 2.4, no location with more than 4 roads. Farthest location from the HQ at least floor(V / 4) roads away. At least 3 waves, and the union of all waves leaves at least one target unreachable from the HQ
6. Outbreak Camp	A relief camp with an outbreak. Medicine and water count and supplies spoil. Radio beacons repeat the same messages	Medicine and water crates (cat 2, 1) at least 55%. At least half of the crates expire by hour 300. One symbol is at least 25% of the log

2. The four trap cases

Cases 46 to 49 are deliberately hostile but complete, valid cases (tier X). Each takes the story of its case study (46: study 5, 47: study 6, 48: study 1, 49: study 2) with its signature rules waived.

Case	Name	Requirement
46	Dock-order trap	1,500 to 3,000 crates. The list, in crate_id order, is already monotone (all non-decreasing or all non-increasing) in at least three of the five fields value, weight, expiry, serial and value-per-kg
47	Best-looking-crate trap	12 to 60 crates. Taking crates in decreasing order of value per kg and skipping those that do not fit reaches **at most 60%** of the best possible total value
48	Shortest-looks-cheapest trap	1,500 to 3,000 locations, with a longest distance of at least 500 roads between two locations. For at least one target, **every** route with the fewest roads costs at least **1.5 times** a cheapest route
49	Near-miss trap	50,000 to 100,000 log symbols, at least 90% of them a repetition of one block of at most 8 symbols. For at least four patterns, the first (length - 1) symbols match the log at no fewer than half of all start positions

3. The Excel workbook (cases.xlsx)

The workbook has the 20 sheets below, in this order. Row 1 of each sheet is the header with exactly these column names. The first column is always **case_id**. Rows are sorted by case_id, then by the second column. A case is spread over several sheets and joined by case_id.

Ranges by tier

Item	Tier S	Tier M	Tier L
Crates	8 to 20	40 to 200	500 to 1,500
Locations V	6 to 12	30 to 150	500 to 2,000
Roads E	V - 1 to 3V	V - 1 to 3V	V - 1 to 3V
Closure waves	2 to 5	2 to 5	2 to 5
Radio log (symbols)	200 to 500	2,000 to 10,000	20,000 to 100,000
Length of each report	10 to 30	30 to 150	200 to 1,000
Windows	8 to 20	50 to 300	1,000 to 3,000
Alarms	10 to 40	200 to 1,000	2,000 to 5,000
Relief sites	5 to 9	10 to 16	200 to 500
Pipeline stages	4 to 8	15 to 60	100 to 300
Drum length	5 to 14	40 to 200	500 to 2,000
Sensor nodes	6 to 14	20 to 40	100 to 400
Safety variables (switches)	6 to 15	20 to 50	20 to 50
Ration quantities	8 to 20	24 to 40	50 to 100

Sheets, columns and rules

Sheet	Columns	Rules
cases	case_id, group_seed, case_no, tier, scenario, capacity, target_0, target_1, target_2, edit_bound, tour_length, sensor_nodes, sensor_k, safety_vars, rations_target	One row per case (50 rows). scenario = (case_no mod 6) + 1. capacity is 10% to 70% of the total crate weight, at least 1. The three targets are distinct shelters or hospitals, not the HQ
crates	case_id, crate_id, cat, weight, value, expiry, serial	crate_id runs 1, 2, 3, ... cat: 0 Food, 1 Water, 2 Medicine, 3 Tool, 4 Rare. weight 1 to 20 kg. value 10 to 100. expiry hour 0 to 999. serial a distinct 8-digit integer (10,000,000 to 99,999,999)
locations	case_id, loc_id, type, penalty	loc_id runs 0 to V - 1. type D (depot), S (shelter), H (hospital) or J (junction). penalty in {0, 1, 2, 4, 6}. Location 0 is the HQ: type D, penalty 0. At least one S and one H
roads	case_id, road_no, u, v, minutes, cable	road_no runs 0, 1, ... u < v, no repeated pair. minutes 1 to 60, cable 1 to 100. The network is connected before any closure. The cost of the step u to v is minutes + penalty of v

Sheet	Columns	Rules
waves	case_id, wave_no, road_no	One row per closed road. Waves are disjoint, and each has 1 to max(1, floor(E / 4)) roads
radio_log	case_id, part, text	Alphabet of 38 symbols: A to Z, 0 to 9, underscore, hyphen. The log is the text cells joined in order of part (1, 2, ...). One cell holds at most 10,000 symbols
radio_patterns	case_id, pattern_no, pattern, count	Exactly six patterns (pattern_no 0 to 5) of lengths 2, 4, 8, 12, 20, 40 in this order. count is the number of occurrences in the log, overlaps included. At least two patterns occur and at least one does not
radio_reports	case_id, report_a, report_b	Two strings over the alphabet. edit_bound (in cases) is at least the true minimum number of insertions, deletions and replacements turning report_a into report_b, and at most len(report_a) + len(report_b)
windows	case_id, window_id, start, finish, value	window_id runs 1, 2, ... 0 <= start < finish <= 1440 (minutes). Duration 5 to 480. value 1 to 100. With 8 or more windows, at least two pairs touch (one finishes when the other starts)
incidents	case_id, alarm_id, tick, priority	alarm_id runs 1, 2, ... tick starts at 0 and never decreases. priority 1 to 10 (10 most urgent). On tiers M and L: alarms per tick (count / (last tick + 1)) between 1.2 and 1.8, and at least 6 distinct priorities including 1
sites	case_id, site_no, x, y	site_no runs 0 to n - 1. x, y integers 0 to 999. No two sites equal
tour	case_id, step, site_no	A planted tour: a permutation of all sites starting at site 0. tour_length (in cases) is its closed Manhattan length. On tiers L and X it is no longer than the tour that always visits the nearest unvisited site from site 0
pipeline_dims	case_id, idx, dim	idx 0 to n; n + 1 integers from 2 to 100 (matrix i is dim(i - 1) by dim(i))
drum_prices	case_id, length, price	Price of a piece of length j for j = 1 to L, a positive integer up to 500. Prices must not all be proportional to length
sensor_edges	case_id, u, v	Graph on nodes 0 to sensor_nodes - 1. u < v, no loops, no repeated edge, at least n - 1 edges
sensor_cover	case_id, node	A planted cover of sensor_k distinct nodes, 2 <= k <= n / 2, touching every edge
safety_rules	case_id, rule_no, lit_1, lit_2, lit_3	Each rule has three non-zero integers on three different switches: +j means switch j ON, -j means OFF. A rule holds if at least one condition holds. Rules divided by safety_vars is between 2.0 and 6.0
safety_planted	case_id, switch, value	Value 0 or 1 for switches 1 to safety_vars that satisfies every rule
rations	case_id, item, value, planted	item runs 0, 1, ... value a positive integer up to 1,000. planted is 1 for the planted selection, 0 otherwise. Planted values add up to rations_target (positive)
claims	case_id, cargo_opt, window_count_opt, window_value_opt, drum_opt, tour_opt, cover_min	**Tier S only.** Exhaustive-search optima: best crate value within capacity; most compatible windows (touching allowed); best compatible window value; best total drum price; shortest closed tour; fewest nodes touching every sensor edge

Variety across your 50 cases

1. In at least 5 tier-S cases, taking crates in decreasing order of value per kg and skipping those that do not fit is **not** optimal.
2. In at least 5 tier-S cases, for at least one target, every route with the fewest roads costs more than a cheapest route.
3. Across your 26 cases of tiers M and L, safety rules per switch lies between 2.5 and 6.0, with at least 5 cases below 3.5, at least 8 from 3.5 to 4.8, and at least 5 above 4.8.
4. No two of your cases are equal in any module of size 8 or more.

4. Validation

Your validator must **recompute** everything and not trust the fields. It must check at least:

#	Check
V1	Exactly the 20 sheets and headers of Section 3; 50 distinct case_ids in the right format; the correct tier and scenario for each case_no

#	Check
V2	Every size and value is inside its range
V3	Identifiers are unique and consecutive where required; every reference (road ends, targets, waves, planted items, cover nodes) exists
V4	The network is connected before closures; waves are disjoint and within their size limit
V5	Every pattern count and edit_bound is correct; log parts are in order and within 10,000 symbols; claims are correct on tier S
V6	Every planted object is valid: the tour and its length, the cover, the assignment, the ration selection
V7	The signature rules of each case (0 to 45) and the variety rules
V8	The trap rules of cases 46 to 49

The validator must also **reject bad data**. Keep your broken copies in validator/tests/ and show their rejection in validation_report.txt.

5. The approach note

Two pages, four parts:

1. **From story to numbers.** For each case study, the main choices you made (sizes, distributions, shapes) and the sentence of the story that justifies each.
2. **Planted objects and claims.** How your generator makes each one correct by construction.
3. **The traps.** For each trap, what you built, why it is hostile, and which validator check covers it.
4. **What will be hard.** For each case study, the two relief decisions you expect to be hardest and the property of your data that makes them so.

6. Algorithm-selection report (two pages)

The group submits a report, also called the proposal, of **exactly two pages, in PDF format**, for the decision modules it will implement in Iteration 2. No algorithm is prescribed: the group recognises the problem behind each decision and selects the algorithms.

Module	Decision	Input	Output and objective
A. Relief inventory	What to load on the truck	crates, capacity, the values of the day	A set of indivisible crates of total weight at most the capacity, with maximum total value
B. Route planning	Which route to drive on today's map	network, source, today's blocked roads	A cheapest route and its cost, or no route; also the set of reachable locations and a fewest-roads route
C. Rescue scheduling	Which windows to staff	windows	A set of pairwise compatible windows (touching windows are compatible) with the most windows, or the most total value
D. Emergency queue	Which alarm to answer first	incidents	The release order, one alarm per tick, in urgency order among the alarms present
E. Radio logs	Phrase search, report comparison, compact transmission	radio	Every start position of a phrase, overlaps included; the minimum number of insertions, deletions and replacements between the two reports, with an edit script; a prefix-free binary code of minimum total length for the log
F. Network planning	How to lay cable	network	A set of roads linking all reachable locations with minimum total cable cost
G. Chains and cuts	How to order signal-processing stages and cut a drum	pipeline	A bracketing of the chain with the fewest scalar multiplications (p-by-q times q-by-r costs p *q* r); a way to cut a drum of length L with maximum total price
H. Hard planning questions	Four questions nobody can answer quickly	safety, rations, tour, sensors	A satisfying assignment of the safety rules, or none; a ration selection summing exactly to the target, or none; a shortest closed tour through all sites; at most k sensor nodes touching every edge, or none
I. Nearest sites and checksums	Closest sites; long numbers	sites, crates	Two sites at minimum squared Euclidean distance; the product of two long decimal numbers built from the serials

The page contains **one table with one row per chosen module** and nothing else. The table has these columns:

Column	Content
Module	Letter and name from the table above
Problem	Input, output and objective of the decision, in the variables used for complexity (n, m, V, E, W)
Candidates	At least two algorithms, each with its time complexity
Selection	The primary algorithm and the baseline for comparison
Justification	The data property and the assumption on which the selection rests, and the case in which it would fail

Page 2: work division. The group decides for itself how the work is shared. The second page contains one table with one row per member, with the columns:

Column	Content
Member	Name, roll number and section
Contribution	What the member did in Iteration 1: data modules, validator checks, claims, approach note, and the decision modules the member proposes for Iteration 2

Every member must be able to explain the whole submission.

7. GitHub repository

The group creates one private GitHub repository at the start of Iteration 1, names it with the same pattern as the ZIP (RollNoSection1_RollNoSection2_RollNoSection3), adds the course staff as collaborators, and keeps all of its work in it. The repository continues into Iteration 2.

- Commit as each part is completed (for example a generator module, a validator check, a draft of the report), with a message that describes the change.
- Each member commits their own work from their own account.
- Commit step by step over the days of the iteration, not in a single bulk commit at the end.

8. What you submit

One ZIP per group, named **RollNoSection1_RollNoSection2_RollNoSection3** (for example 22I-3105-A_22I-3106-B_22I-3107-C), containing:

Item	What it must contain
cases.xlsx	The Excel workbook with exactly 50 cases, in the format of Section 3. Written by your generator, never edited by hand
generator/	Your own program that writes cases.xlsx from the group seed, with a README giving the one command that reproduces the file
validator/	Your own program that checks cases.xlsx against Section 4 and prints PASS or the broken rules for every case. Also validator/tests/ with at least two valid cases, each broken in at least five different ways
validation_report.txt	Output of your validator: PASS for all 50 cases, and the rejection of every broken copy
approach note	2 pages (Word or PDF) with the four parts of Section 5
proposal.pdf	The two-page algorithm-selection report (proposal) and work division of Section 6, **in PDF format**
repository.txt	The link to the group's GitHub repository (Section 7)
members.txt	Three names, roll numbers and sections, and the group number

Good luck.